Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	Ⅰ -Ⅲ Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
Student Name	Keerthi Priya Ande
Roll Number	2520090177
Branch	CSIT
Course Outcome	2

Case Study Topic: CampusFlow – University Campus Resource Management System using B+ Tree

Work Done Summary:

Implemented a B+ Tree in Java for efficient campus resource storage and retrieval. Stored classroom, laboratory, and facility records in the B+ Tree for fast searching and indexing operations. Inserted multiple campus resource records and supported dynamic updates during scheduling and resource bookings. Implemented efficient search operations to retrieve facility information based on department and resource IDs. Supported fast access, sorted data storage, and infrastructure optimization using B+ Tree properties. B+ Tree provides efficient performance for large-scale campus database indexing and real-time resource management systems.

Implementation Details :

1. Defined B+Tree class with node structures for internal and leaf nodes.
2. Inserted campus resource records such as classrooms, labs, and faculty allocations into the B+ Tree.
3. Implemented search operations to retrieve resource details efficiently using resource IDs.
4. Leaf nodes store actual resource data and are linked for sequential access.

5. Internal nodes maintain indexing keys for fast traversal and searching.
6. The main method inserted resource records, performed searches, and displayed stored campus data.
7. B+ Tree supports efficient indexing and storage management for large-scale university systems.

Result : Screenshots/Output:

```
Original Campus Resources:
Resource ID 101 = Computer Lab
Resource ID 102 = Seminar Hall
Resource ID 103 = Library
Resource ID 104 = Physics Lab
Resource ID 105 = Classroom A
Resource ID 106 = Classroom B

B+ Tree Leaf Nodes:
Key = 101
Key = 102
Key = 103
Key = 104
Key = 105
Key = 106

Total Campus Resources = 6

Searching Resource 104...

Resource Found:
Resource ID 104 = Physics Lab

Searching Resource 110...

Resource Not Found

B+ Tree Indexing Completed Successfully.
```

Time Complexity:

- Insertion Operation: **O(log n)**
- Search Operation: **O(log n)**

- Deletion Operation: **O(log n)**
- B+ Tree is highly efficient for database indexing and campus resource management systems.

GitHub Link :https://github.com/keerthipriya2008/DSA2_CO2

Student Signature	**Faculty Signature**